In [ ]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set visual style for plots          Pragya singhal , CU24220050
sns.set_theme(style="whitegrid")

# Step 1 & 2: Load the Netflix dataset
file_path = 'netflix_titles.csv'
try:
    df = pd.read_csv(file_path)
    print("Dataset loaded successfully!")
except FileNotFoundError:
    print(f"Error: '{file_path}' not found. Please ensure the file is in your wo

# Step 3: Explore structure, data types, dimensions, and summary statistics   Pr
print("\n--- Dataset Dimensions (Rows, Columns) ---")
print(df.shape)

print("\n--- Dataset Info & Data Types ---")
print(df.info())

print("\n--- Missing Values Check ---")
print(df.isnull().sum())

print("\n--- Summary Statistics (Categorical & Numerical) ---")
print(df.describe(include='all'))

# Step 4: Analyze distribution of numerical features (e.g., Release Year)   Prag
plt.figure(figsize=(10, 5))
sns.histplot(df['release_year'], bins=30, kde=True, color='purple')
plt.title('Distribution of Release Years on Netflix', fontsize=14)
plt.xlabel('Release Year')
plt.ylabel('Count')
plt.show()

# Step 5: Visualize categorical variables (Movies vs TV Shows & Top Ratings)
plt.figure(figsize=(14, 5))

plt.subplot(1, 2, 1)
sns.countplot(data=df, x='type', palette='Set2')
plt.title('Count of Movies vs. TV Shows', fontsize=12)

plt.subplot(1, 2, 2)
sns.countplot(data=df, y='rating', order=df['rating'].value_counts().index, pale
plt.title('Distribution of Content Ratings', fontsize=12)

plt.tight_layout()
plt.show()

# Step 6 & 7: Top Countries and Genre Analysis (Bivariate / Categorical Analysis
plt.figure(figsize=(12, 5))
top_countries = df['country'].value_counts().head(10)
sns.barplot(x=top_countries.values, y=top_countries.index, palette='coolwarm')
plt.title('Top 10 Countries Producing Content on Netflix', fontsize=14)
plt.xlabel('Number of Titles')
plt.ylabel('Country')
```

```
plt.show()

# Step 8: Multivariate Analysis (Content additions over the years by Type)
if 'date_added' in df.columns:
    # Extract year from date_added
    df['year_added'] = pd.to_datetime(df['date_added'].str.strip(), errors='coer

    plt.figure(figsize=(12, 6))
    sns.countplot(data=df, x='year_added', hue='type', palette='Set1')
    plt.title('Content Additions Over the Years (Movies vs TV Shows)', fontsize=
    plt.xlabel('Year Added to Netflix')
    plt.ylabel('Number of Titles')
    plt.xticks(rotation=45)
    plt.legend(title='Type')
    plt.show()

# Step 9 & 10: Summary of Findings (Printed Output)   Pragya singhal , CU2422005
print("\n--- Key Business Insights Summary ---")
print(f"Total Titles Analyzed: {df.shape[0]}")
print(f"Total Movies vs TV Shows breakdown:\n{df['type'].value_counts()}")
print(f"Top Producing Country: {df['country'].mode()[0]}")
print(f"Most Common Rating: {df['rating'].mode()[0]}")
```

```
Dataset loaded successfully!

--- Dataset Dimensions (Rows, Columns) ---
(8807, 12)

--- Dataset Info & Data Types ---
<class 'pandas.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   str
 1   type          8807 non-null   str
 2   title         8807 non-null   str
 3   director      6173 non-null   str
 4   cast          7982 non-null   str
 5   country       7976 non-null   str
 6   date_added    8797 non-null   str
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   str
 9   duration      8804 non-null   str
 10  listed_in     8807 non-null   str
 11  description   8807 non-null   str
dtypes: int64(1), str(11)
memory usage: 825.8 KB
None

--- Missing Values Check ---
show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        10
release_year       0
rating             4
duration           3
listed_in          0
description        0
dtype: int64

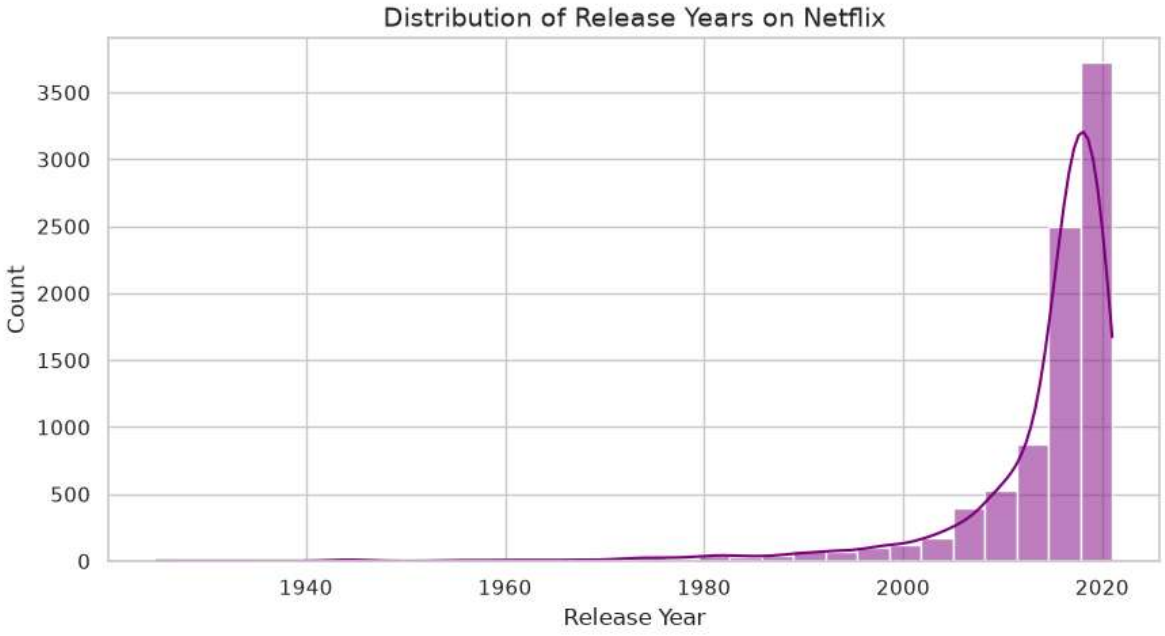
--- Summary Statistics (Categorical & Numerical) ---
       show_id   type                  title       director  \
count     8807   8807                   8807           6173
unique    8807      2                   8807           4528
top         s1  Movie  Dick Johnson Is Dead  Rajiv Chilaka
freq         1   6131                      1             19
mean       NaN    NaN                    NaN            NaN
std        NaN    NaN                    NaN            NaN
min        NaN    NaN                    NaN            NaN
25%        NaN    NaN                    NaN            NaN
50%        NaN    NaN                    NaN            NaN
75%        NaN    NaN                    NaN            NaN
max        NaN    NaN                    NaN            NaN

                       cast        country       date_added  release_year  \
count                  7982           7976             8797   8807.000000
unique                 7692            748             1767           NaN
top     David Attenborough  United States  January 1, 2020           NaN
```

freq	19	2818	109	NaN
mean	NaN	NaN	NaN	2014.180198
std	NaN	NaN	NaN	8.819312
min	NaN	NaN	NaN	1925.000000
25%	NaN	NaN	NaN	2013.000000
50%	NaN	NaN	NaN	2017.000000
75%	NaN	NaN	NaN	2019.000000
max	NaN	NaN	NaN	2021.000000

	rating	duration	listed_in	\
count	8803	8804	8807	
unique	17	220	514	
top	TV-MA	1 Season	Dramas, International Movies	
freq	3207	1793	362	
mean	NaN	NaN	NaN	
std	NaN	NaN	NaN	
min	NaN	NaN	NaN	
25%	NaN	NaN	NaN	
50%	NaN	NaN	NaN	
75%	NaN	NaN	NaN	
max	NaN	NaN	NaN	

	description
count	8807
unique	8775
top	Paranormal activity at a lush, abandoned prope...
freq	4
mean	NaN
std	NaN
min	NaN
25%	NaN
50%	NaN
75%	NaN
max	NaN



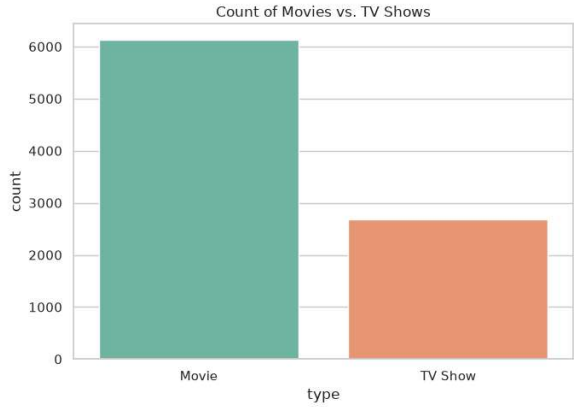
```
/tmp/ipykernel_9463/776432219.py:42: FutureWarning:

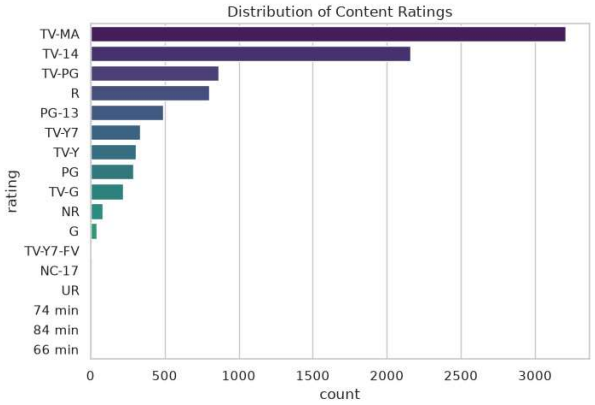
Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.countplot(data=df, x='type', palette='Set2')
/tmp/ipykernel_9463/776432219.py:46: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.countplot(data=df, y='rating', order=df['rating'].value_counts().index, pal
ette='viridis')
```

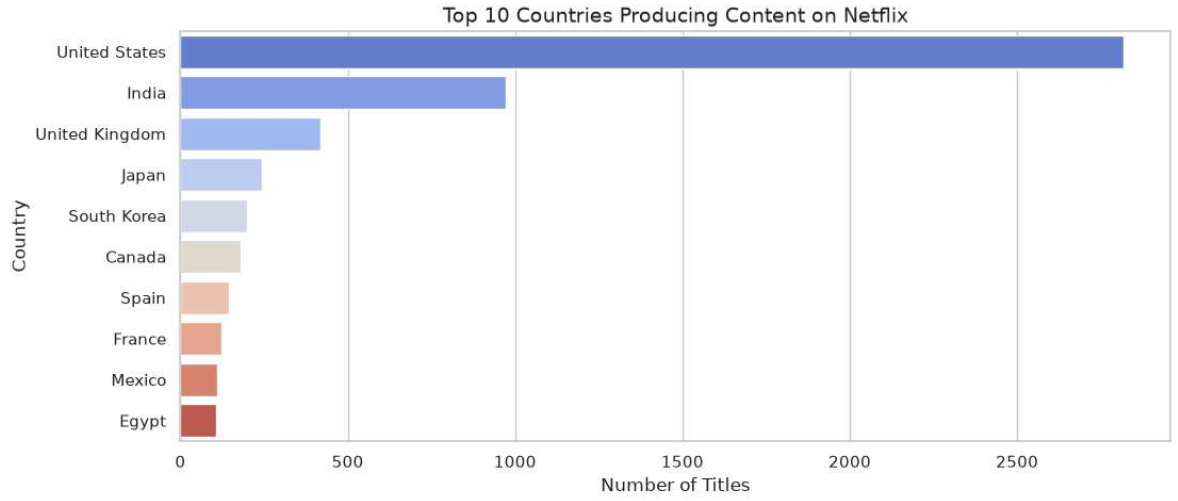


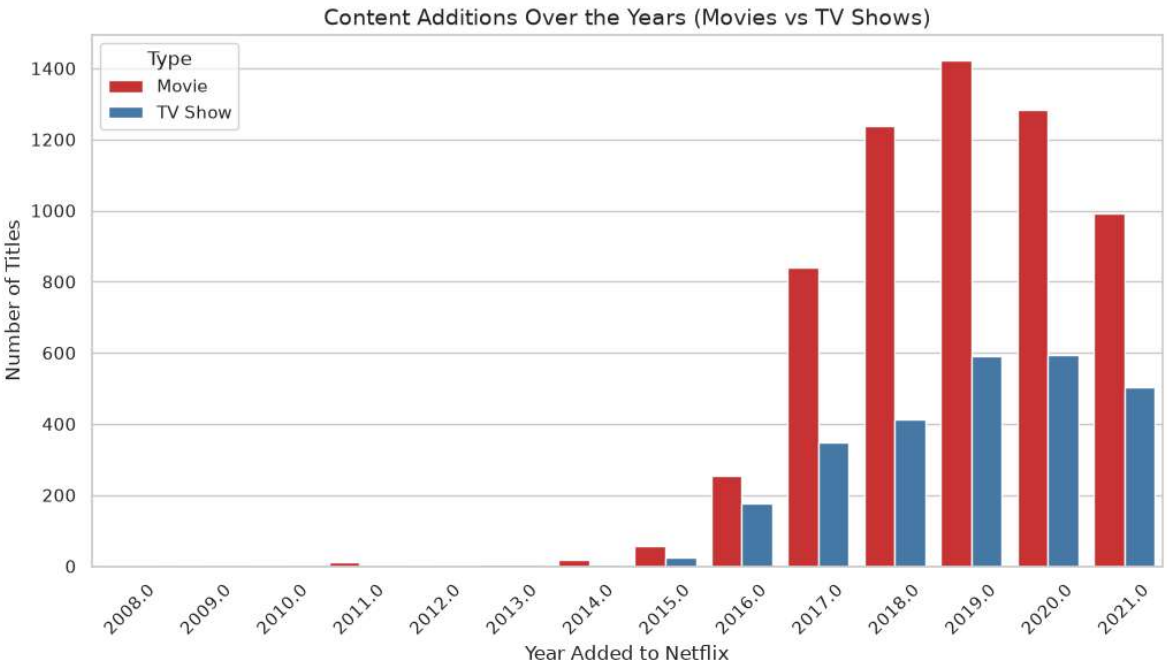



```
/tmp/ipykernel_9463/776432219.py:55: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.barplot(x=top_countries.values, y=top_countries.index, palette='coolwarm')
```






```
--- Key Business Insights Summary ---
Total Titles Analyzed: 8807
Total Movies vs TV Shows breakdown:
type
Movie      6131
TV Show    2676
Name: count, dtype: int64
Top Producing Country: United States
Most Common Rating: TV-MA
```